

Orbital Debris Mitigation Handbook

The growing population of non-functional spacecraft in low Earth orbit presents a cumulative hazard to every mission sharing that altitude band. This handbook describes the disposal practices adopted by flight teams.

Mandatory passivation is required within five years of end of life. Operators must command the satellite to discharge all stored energy, vent residual pressurants, and inhibit all rotating mechanisms.

For satellites operating below 600 km, a dedicated disposal manoeuvre is impractical; operators rely on natural atmospheric drag to remove the vehicle. Debris density is highest between 700 and 1000 km, where drag is weakest and objects can persist for centuries.

Attitude Control Subsystem Overview

The attitude control subsystem is responsible for maintaining the commanded orientation of the spacecraft and for rejecting disturbances from environmental torques.

Three-wheeled reaction wheel assemblies are the standard non-propulsive method. Each wheel stores angular momentum and exchanges it with the body, producing torque without consuming propellant. Wheels saturate when a manoeuvre demands more momentum than the stored capacity.

When wheel momentum is exhausted, thrusters must be fired. A desaturation manoeuvre trades propellant for wheel capacity and is budgeted against the end-of-life propellant margin.

Reaction wheels are highly sensitive to bearing friction. Friction builds with spin rate and temperature, and a degraded wheel can introduce coupling between control axes that must be compensated in the control law.

Ground Station Operations and Scheduling

The ground segment allocates station time to spacecraft according to a published schedule. Each pass is characterised by acquisition of signal, a tracking arc, and loss of signal.

Data volume recorded aboard the spacecraft determines how much of the tracking arc is spent on playback rather than real-time science. A high-rate instrument may require contact durations an order of magnitude longer than a beacon-only orbit.

Scheduling must satisfy pointing, illumination and elevation constraints simultaneously. A station is usable only when the spacecraft is above the minimum elevation of the antenna and the sun is sufficiently far from the boresight to protect the receiver front end.